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18.02 Multivariable Calculus
Fall 2007

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18.02 Lecture 25. – Fri, Nov 9, 2007

Handouts: Exam 3 solutions.

Triple integrals: $\iiint_R f dV$ (dV = volume element).

Example 1: region between paraboloids $z = x^2 + y^2$ and $z = 4 - x^2 - y^2$ (picture drawn), e.g. volume of this region: $\iiint_R 1 dV = \int_{\text{?}}^{\text{?}} \int_{\text{?}}^{\text{?}} \int_{x^2+y^2}^{4-x^2-y^2} dz dy dx$.

To set up bounds, (1) for fixed (x, y) find bounds for z : here lower limit is $z = x^2 + y^2$, upper limit is $z = 4 - x^2 - y^2$; (2) find the shadow of R onto the xy -plane, i.e. set of values of (x, y) above which region lies. Here: R is widest at intersection of paraboloids, which is in plane $z = 2$; general method: for which (x, y) is z on top surface $>$ z on bottom surface? Answer: when $4 - x^2 - y^2 > x^2 - y^2$, i.e. $x^2 + y^2 < 2$. So we integrate over a disk of radius $\sqrt{2}$ in the xy -plane. By usual method to set up double integrals, we finally get:

$$V = \int_{-\sqrt{2}}^{\sqrt{2}} \int_{-\sqrt{2-x^2}}^{\sqrt{2-x^2}} \int_{x^2+y^2}^{4-x^2-y^2} dz dy dx.$$

Evaluation would be easier if we used polar coordinates $x = r \cos \theta$, $y = r \sin \theta$, $x^2 + y^2 = r^2$: then

$$V = \int_0^{2\pi} \int_0^{\sqrt{2}} \int_{r^2}^{4-r^2} dz r dr d\theta.$$

(evaluation easy, not done).

Cylindrical coordinates. (r, θ, z) , $x = r \cos \theta$, $y = r \sin \theta$. r measures distance from z -axis, θ measures angle from xz -plane (picture shown).

Cylinder of radius a centered on z -axis is $r = a$ (drawn); $\theta = 0$ is a vertical half-plane (not drawn).

Volume element: in rect. coords., $dV = dx dy dz$; in cylindrical coords., $dV = r dr d\theta dz$. In both cases this is justified by considering a small box with height Δz and base area ΔA , then volume is $\Delta V = \Delta A \Delta z$.

Applications: Mass: $M = \iiint_R \delta dV$.

Average value of f over R : $\bar{f} = \frac{1}{Vol} \iiint_R f dV$; weighted average: $\bar{f} = \frac{1}{Mass} \iiint_R f \delta dV$.

In particular, center of mass: $(\bar{x}, \bar{y}, \bar{z})$ where $\bar{x} = \frac{1}{Mass} \iiint_R x \delta dV$.

(Note: can sometimes avoid calculation using symmetry, e.g. in above example $\bar{x} = \bar{y} = 0$).

Moment of inertia around an axis: $I = \iiint_R (\text{distance from axis})^2 \delta dV$.

About z -axis: $I_z = \iiint_R r^2 \delta dV = \iiint_R (x^2 + y^2) \delta dV$. (consistent with I_0 in 2D case)

Similarly, about x and y axes: $I_x = \iiint_R (y^2 + z^2) \delta dV$, $I_y = \iiint_R (x^2 + z^2) \delta dV$
(setting $z = 0$, this is consistent with previous definitions of I_x and I_y for plane regions).

Example 2: moment of inertia I_z of solid cone between $z = ar$ and $z = b$ ($\delta = 1$) (picture drawn):

$$I_z = \iiint_R r^2 dV = \int_0^b \int_0^{2\pi} \int_0^{z/a} r^2 r dr d\theta dz \quad \left(= \frac{\pi b^5}{10a^4} \right).$$

(I explained how to find bounds in order $dr d\theta dz$: first we fix z , then slice for given z is the disk bounded by $r = z/a$; the first slice is $z = 0$, the last one is $z = b$).

Example 3: volume of region where $z > 1 - y$ and $x^2 + y^2 + z^2 < 1$? Pictures drawn: in space, slice by yz -plane, and projection to xy -plane.

The bottom surface is the plane $z = 1 - y$, the upper one is the sphere $z = \sqrt{1 - x^2 - y^2}$. So inner is $\int_{1-y}^{\sqrt{1-x^2-y^2}} dz$. The shadow on the xy -plane = points where $1 - y < \sqrt{1 - x^2 - y^2}$, i.e. squaring both sides, $(1 - y)^2 < 1 - x^2 - y^2$ i.e. $x^2 < 2y - 2y^2$, i.e. $-\sqrt{2y - 2y^2} < x < \sqrt{2y - 2y^2}$. So we get:

$$\int_0^1 \int_{-\sqrt{2y-2y^2}}^{\sqrt{2y-2y^2}} \int_{1-y}^{\sqrt{1-x^2-y^2}} dz dx dy.$$

Bounds for y : either by observing that $x^2 < 2y - y^2$ has solutions iff $2y - y^2 > 0$, i.e. $0 < y < 1$, or by looking at picture where clearly leftmost point is on z -axis ($y = 0$) and rightmost point is at $y = 1$.